

Investigation: Function Calls

Pre-Work

1. Consider the following function call:

`float(15)`

- What is the name of the function? `float`
- How many arguments are being passed? `1`
- What is the return value? `15.0`

2. Consider the following function call:

`int(15.0)`

- What is the name of the function? `int`
- How many arguments are being passed? `1`
- What is the return value? `15`

3. Consider the following function call:

`str(15 + 16.0)`

- What is the name of the function? `str`
- What character separates multiple arguments in a function call? `comma`
- How many arguments are being passed? `1`
- What is the return value? `'31.0'`

4. Consider the following function call:

`len('Hello, world')`

- What is the name of the function? `len`
- How many arguments are being passed? `1`
- What is the return value? `12`
- What data type is this? `int`

5. Consider the following function call:

`round(17.76)`

- What is the name of the function? `round`
- How many arguments are being passed? `1`
- What is the return value? `18`
- What data type is this? `int`

6. Consider the following function call:

`round(17.76, 1)`

- What is the name of the function? `round`
- How many arguments are being passed? `2`
- What is the return value? `17.8`
- What data type is this? `float`

Part A

Partner A should read the prompts. Partner B should type the code into the Python interactive shell, and Partner A should record the results.

Type the following prompts into the Python interactive shell and record the results exactly as they're shown.

15	15
spam = 42	
spam	42
spam + 15	57
spam	42
spam = spam + 15	
spam	57
str(spam)	'57'
spam	57
spam = str(spam)	
spam	'57'
len(spam)	2
float(spam)	57.0
len(spam)	2
len(float(spam))	error
round(42.0)	42
round(42.0, 0)	42.0
round(42.0, 1)	42.0
round(42.0, 5)	42.0

Part B

Switch partners!

<code>import math</code>	
<code>math.pi</code>	<code>3.141592653589793</code>
<code>round(math.pi)</code>	<code>3</code>
<code>round(math.pi, 2)</code>	<code>3.14</code>
<code>round(math.pi, 0)</code>	<code>3.0</code>
<code>pi_str = str(math.pi)</code>	
<code>pi_str</code>	<code>'3.141592653589793'</code>
<code>len(pi_str)</code>	<code>17</code>
<code>len('Hello, ')</code>	<code>7</code>
<code>len('world!')</code>	<code>6</code>
<code>len('Hello, ' + 'world!')</code>	<code>13</code>
<code>hello = 'Hello, '</code>	
<code>world = 'world!'</code>	
<code>hello + world</code>	<code>'Hello, world!'</code>
<code>len(hello)</code>	<code>7</code>
<code>len(world)</code>	<code>6</code>
<code>len(hello + world)</code>	<code>13</code>
<code>len(hello) + len(world)</code>	<code>13</code>
<code>round(math.pi, len(hello))</code>	<code>3.1415927</code>
<code>len(str(round(math.pi, len(hello))))</code>	<code>9</code>